Fundamentals of Electromagnetism

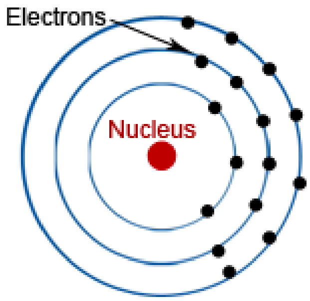
- **Electrostatics:** To study of phenomena associated with electric charge at rest

- **Electrodynamics:** The study of phenomena associated with moving charges

→ Moving electric charges lead to magnetic fields

Electromagnetism:

The study of phenomena associated with moving charges and generated magnetic fields.



Loose an e- ⇒ atom is Positively charged

Gain and e- ⇒ atom is Negatively charged

However charge is conserved within an isolated system, which means that charge cannot be created or destroyed, but can be transferred from one object to another.

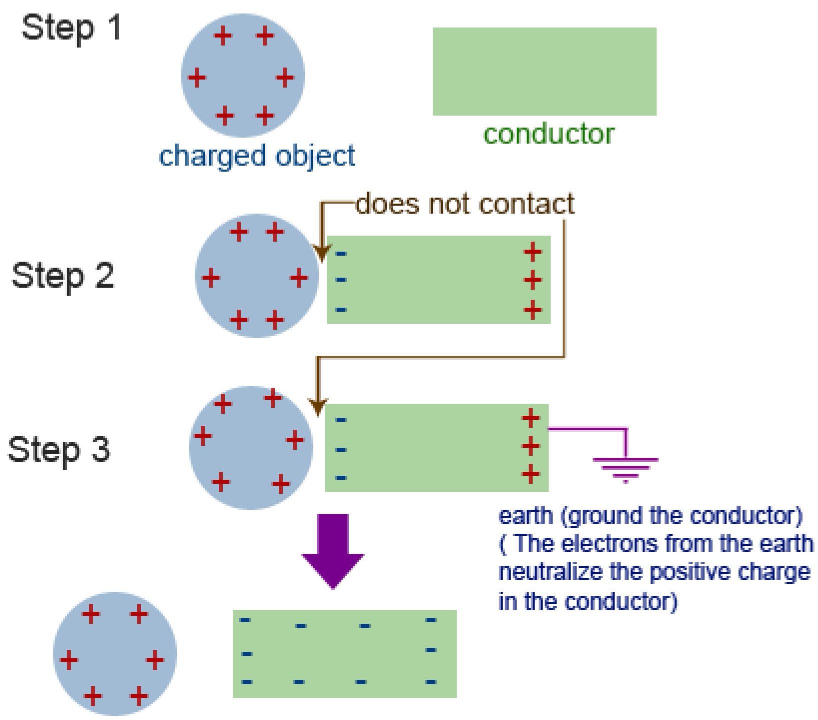
Eg. Rubbing a rod with fur- electrons transfer from fur to rod, leaving rod negatively charged, and the fur with exactly same magnitude of positive charge.

How can we charge an object?

1) By rubbing two insulators

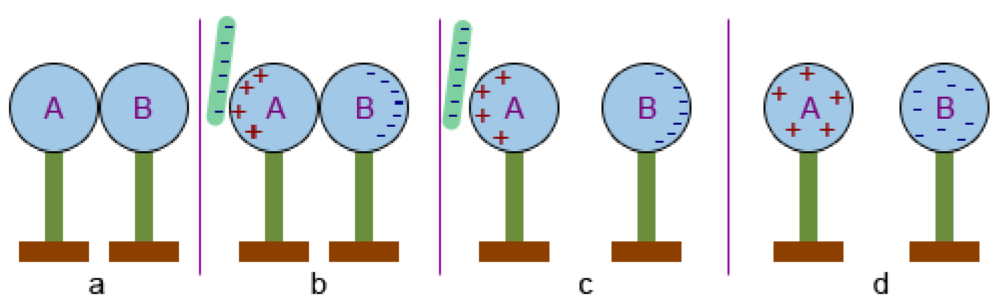
Eg. If a glass rod is rubbed by a piece of silk then the glass rod becomes positively charged and the silk is negatively charged.

2) By induction



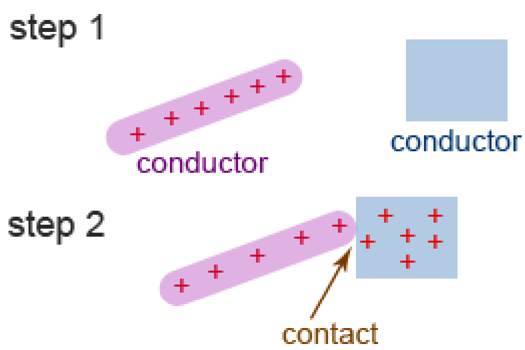
• We can produce a very high charge by repeating the above steps
• Amount of charge acquired > amount of charge in the charged object

Eg:



By induction: Here, in (b), e's in A-B repelled away from rod, so get excess on B, leaving A positively charged
Note, the charged rod never touched them, and retains its original charge

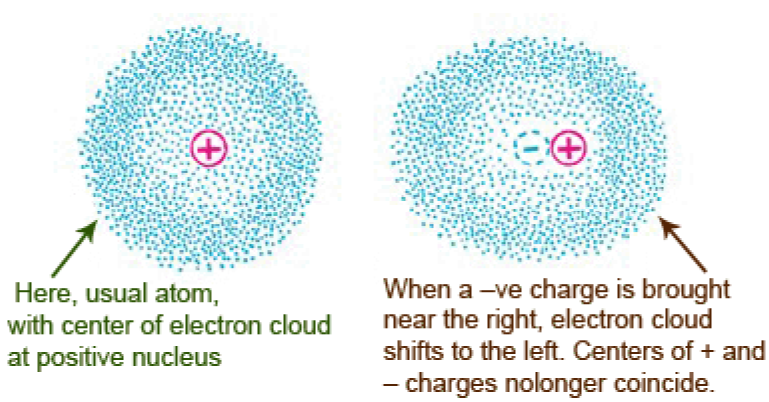
3) By contact



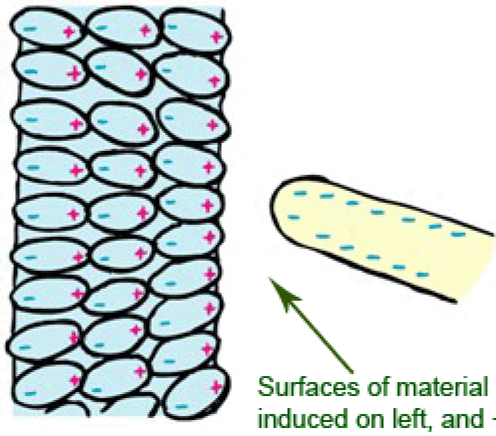
Amount of charge acquired < amount of charge in the charged object.

Charge polarization

When bring a charged object near an insulator, electrons are not free to migrate throughout material. Instead, they redistribute **within** the atoms/molecules themselves: their "centers of charge" move



Atom is ***electrically polarized***

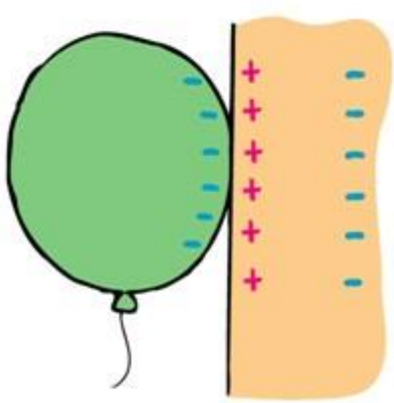


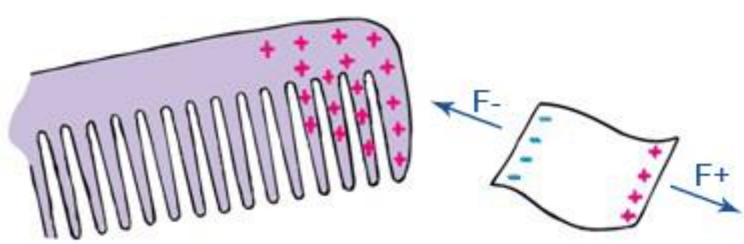
Surfaces of material look like this. A – charge induced on left, and + on the right.
(Zero net charge on whole object)

Charge polarization is why **a charged object can attract a *neutral one*** :

Rub balloon on your hair it will then stick to the wall !
Why?

Balloon becomes charged (by friction) when rub on hair, picking up electrons. It then induces opposite charge on the wall's surface closest to it (Positivly), and the same charge as itself (Negativly) on side of wall furthest away.
So balloon is attracted to positive charges and repelled by negative charges in wall - but the negative charges are further away so repulsive force is weaker and attraction wins.
(Argument applies generally. Key thing is difference in distance between positive and negative)

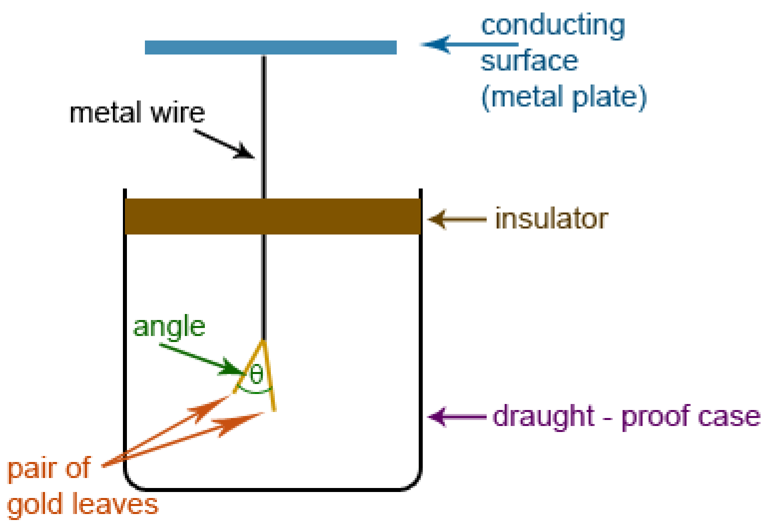




Charge a comb by rubbing it through your hair, and then see it attracts bits of paper and fluff...

Detection of electric charge

• Electroscope is used for this purpose
• The commonest electroscope is the **Gold Leaf Electroscope**



• When a charged object is brought close to the conducting surface the leaves separate owing to their mutual repulsion.
• One of the gold leaves maybe replaced by a vertical metal plate
• The angle between two plates (θ) is proportional to the charge of the object
$\theta \propto$ charge

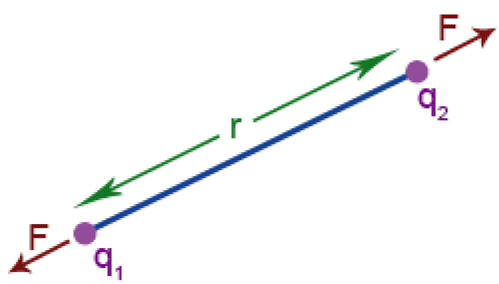
Coulomb's Law:

The mutual electrostatic force between two point charges is directed along the line between charges. The force is directly proportional to the product of the charges and inversely proportional to the square of the distance between them. The force is repulsive if the charges have the same sign and attractive if they have opposite sign.

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = k\frac{q_1 q_2}{r^2} \quad \text{where } k = \frac{1}{4\pi\varepsilon_0}$$

ε_0 - Permittivity of vacuum



• If the charges are in a medium
Permittivity of medium (ε) = ε_r ε_0
ε_r - relative permittivity (dielectric constant)
ε_r varies from unity (1) (for vacuum) to over 4000 (for ferroelectrics), but normally does not exceed 10.

Exercise1: Compare the Coulomb's law with Newton's Gravitational Law

Question 1:

Compare the gravitational force between an electron and proton in an H atom with the electrical force between them. Use:

Average radius of H atom = 0.5 x 10-10 m
Mass of proton = 1.67 x 10-27 kg
Mass of electron = proton mass/2000 = 8.35 x 10-31 kg

Question 2.

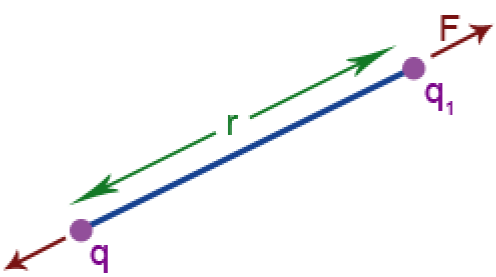
So the electrical attraction is by far dominant in providing the centripetal force that keeps the electron in orbit around the proton. How about the force the electron exerts on the proton – is that larger, same or less than this?

Electric Field:

The space surrounding an electric charge within which it is capable of exerting a perceptible force on another electric charge.

Electric Field Intensity:

The force acting on an unit (Positive) charge placed in a point is the electric field intensity of that point. (The direction of the field is taken to be the direction of the force it would exert on a positive unit charge)



- Electric field intensity (Vector quantity)
- Electric field strength (magnitude of the above vector)

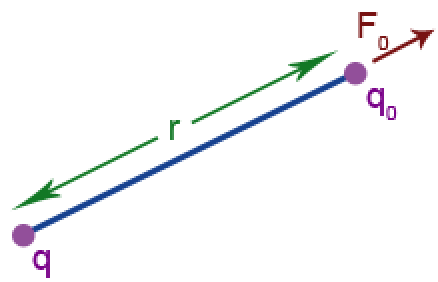
$$F = \frac{1}{4\pi\varepsilon_0}\frac{qq_1}{r^2}$$ Coulomb's Law

if $q_1 = 1$ (unit charge)

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} = E \text{ (Electric field intensity)}$$

Force acting on a point charge (of q_0)

$$F_0 = q_0 E$$

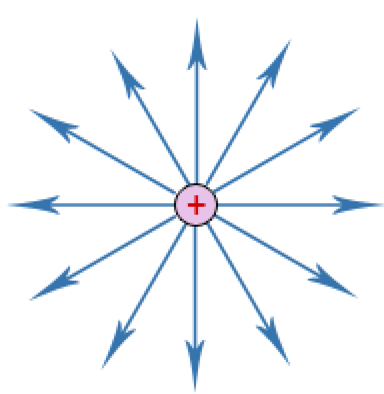


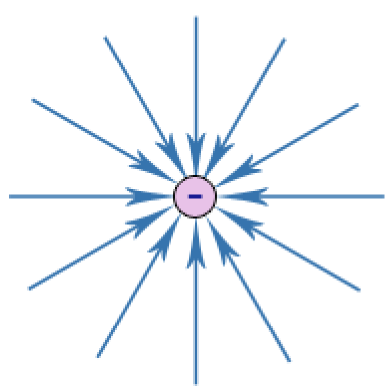
Electric field lines:

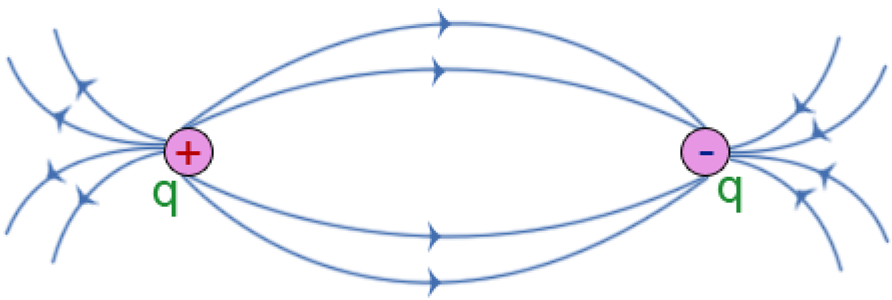
To represent electric field we use imaginary lines called field lines.

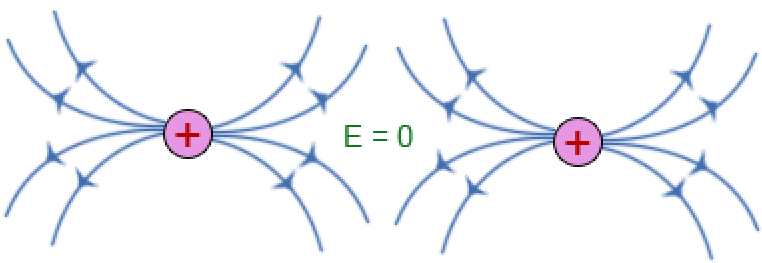
Properties of field lines:
1. Field line start from positive charges and ends at negative charges.
2. Field lines do not intersect each other

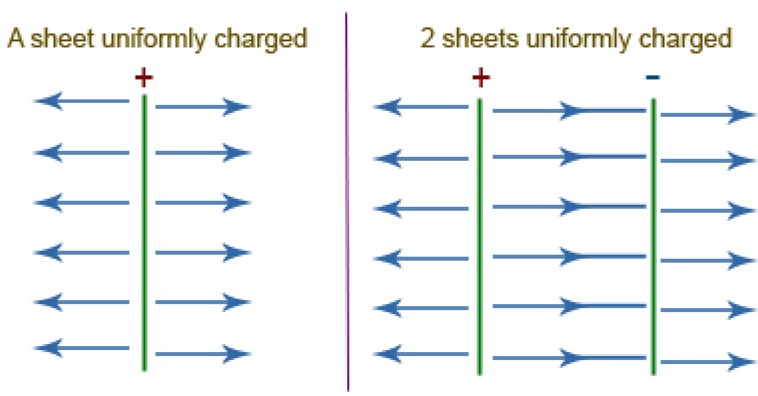
Field for charge configurations:



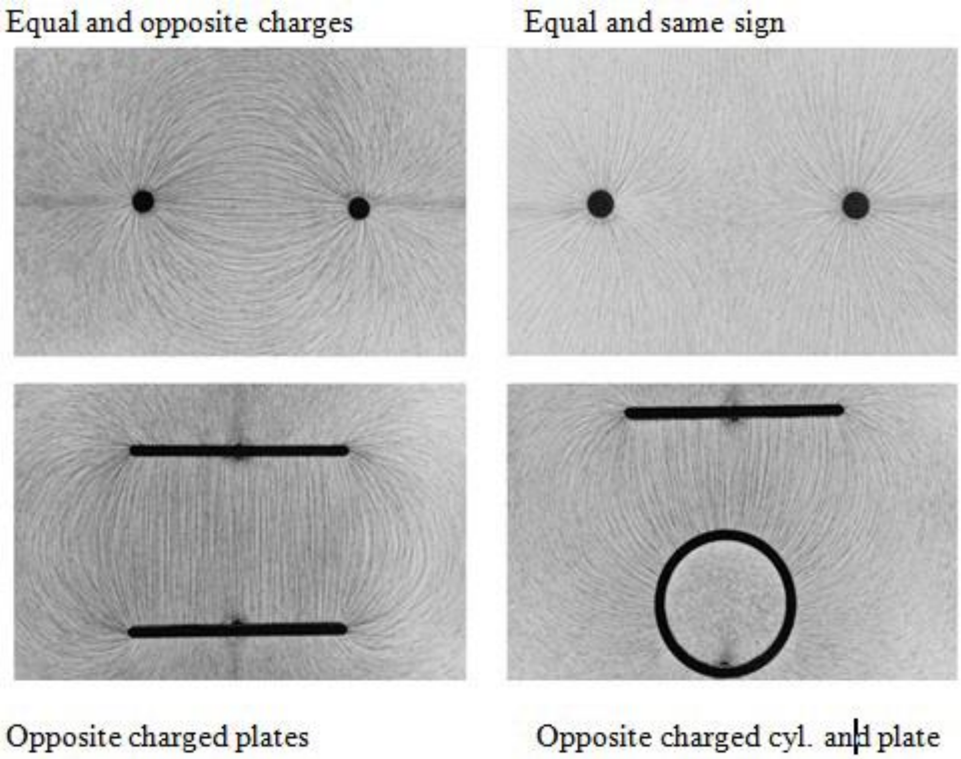






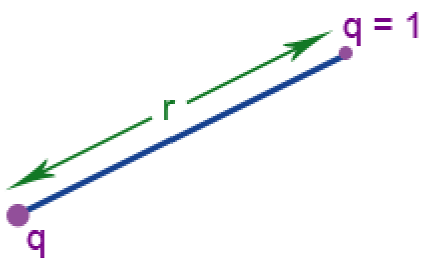


Eg. Field lines shown by small pieces of thread in an oil bath surrounding charged objects:



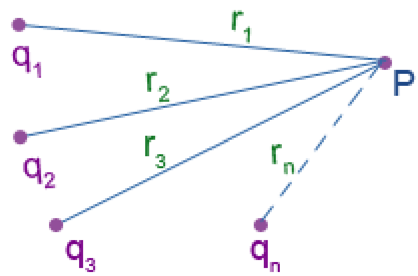
Determination of Electric Field Intensity

Point charges:



$$E = \frac{q}{4\pi\varepsilon_0 r^2}$$

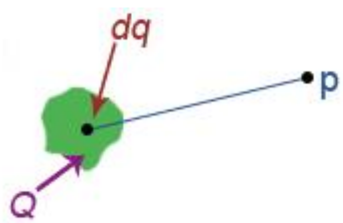
If there are more than one charge



$E = E_1 + E_2 + E_3 + ... + E_n$
Should get the vector sum

Continuous Charge Distribution

Consider a continuous charge (of Q) and take an infinitesimal charge (of dq)



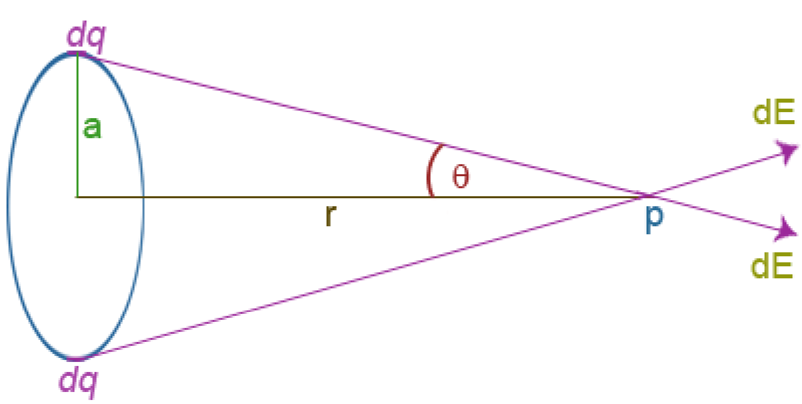
$$dE = \frac{dq}{4\pi\varepsilon_0 r^2}\hat{r} \leftarrow \text{unit vector along } r$$

Electric field intensity
At P due to dq

Electric field intensity
At P due to Q

$$\underline{E} = \int \frac{\hat{r}}{4\pi\varepsilon_0 r^2} dq$$

Eg (1) Ring of charge Q



Consider two infinitesimal charge elements (Directly Opposite) each of *dq*

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{(a^2+r^2)}$$

• Perpendicular components cancel each other
• Parallel components add ⇒ *2dE cos θ*

The electric field intensity at due to these two elements = *2dE cos θ*

$$\Rightarrow E \text{ due to whole ring} = \int 2dE\cos\theta$$

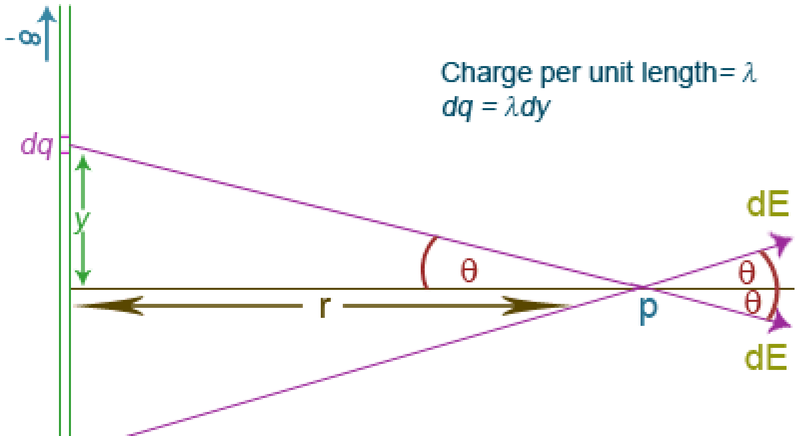
$$E = \int 2\frac{1}{4\pi\varepsilon_0}\frac{dq}{(a^2+r^2)}\frac{r}{(a^2+r^2)^{1/2}}$$

$$E = \frac{r}{2\pi\varepsilon_0(a^2+r^2)^{3/2}}\int dq$$

$$E = \frac{Qr}{4\pi\varepsilon_0(a^2+r^2)^{3/2}}$$

- E at center $= 0$ ($\because r = 0$)
- E at a point considerably far away $= \dfrac{Qr}{4\pi\varepsilon_0 r^3} = \dfrac{Q}{4\pi\varepsilon_0 r^2}$ $[\because (a^2+r^2)^{3/2} \approx r^3]$

Eg (2) Long uniformly charged line

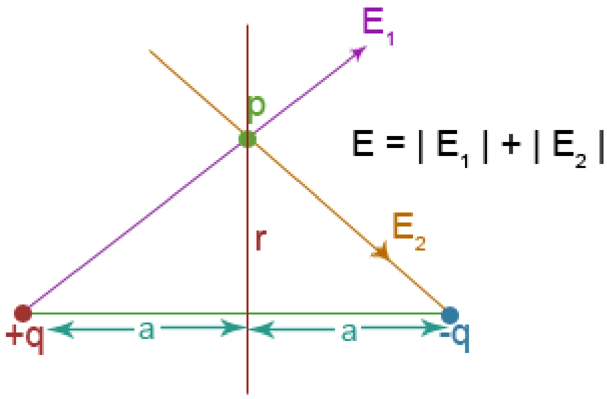


$$dE = \frac{\lambda dy}{4\pi\varepsilon_0 (y^2 + r^2)}$$

$$\begin{aligned}
&E \text{ at } P \text{ due to two infinitesimal charge elements of charge } dq &&= 2dE\cos\theta \\
& &&= \frac{2\lambda dy\cos\theta}{4\pi\varepsilon_0 (y^2 + r^2)} \\
&E \text{ at } P \text{ due to whole line} &&= \frac{\lambda}{2\pi\varepsilon_0}\int_0^{\pi/2} \frac{r d\theta\cos\theta}{(y^2 + r^2)\cos^2\theta} \\
& &&= \frac{\lambda}{2\pi\varepsilon_0 r}\int_0^{\pi/2} \cos\theta d\theta \\
& &&= \frac{\lambda}{2\pi\varepsilon_0 r}[\sin\ \theta]_0^{\pi/2} = \underline{\underline{\frac{\lambda}{2\pi\varepsilon_0 r}}}
\end{aligned}$$

Eg (3) Dipole

A system of two equal and opposite charges placed at a very short distance apart.



$$E = \frac{q}{4\pi\varepsilon_0(a^2 + r^2)}$$

Electric field intensity at P, $= 2E\cos\theta$

$$= \frac{2q}{4\pi\varepsilon_0(a^2 + r^2)} \frac{a}{(a^2 + r^2)^{1/2}}$$

$$= \frac{2qa}{4\pi\varepsilon_0(a^2 + r^2)^{3/2}}$$

Electric dipole moment (**P**):
The product of either of the charges and distance between them
i.e. P = $q(2a)$

Therefore Electric field intensity at P, $E = \frac{\mathbf{P}}{4\pi\varepsilon_0(a^2 + r^2)^{3/2}}$

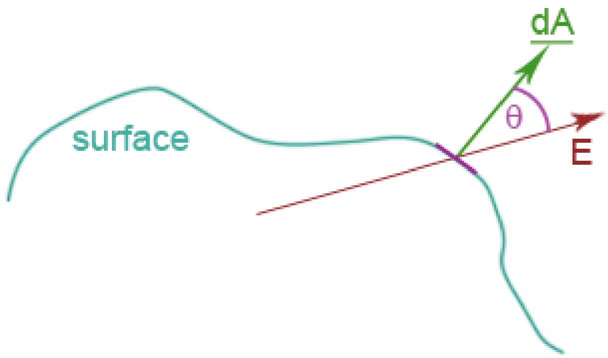
if r >>> a

$$E = \frac{P}{\underline{\underline{4\pi\varepsilon_0 r^3}}} \quad \left.\right\} \quad \begin{array}{l} E_{\text{point charge}} \propto 1/r^2 \\ E_{\text{dipole}} \propto 1/r^3 \end{array}$$

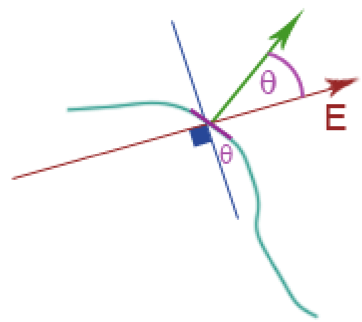
Electric Flux (Φ_E):

Electric flux is a measure of the number of electric field lines <u>crossing the surface</u>

perpendicularly.



$\underline{dA}$ is perpendicular to small area dA



Electric flux crossing the small area $d\phi_E = E\cos\theta dA$

$$\Rightarrow \quad d\phi_E = \underline{E}.\underline{dA}$$

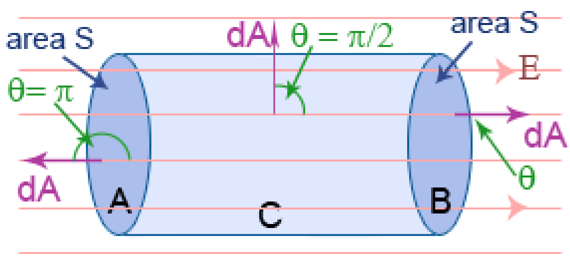
If it is a closed surface $\phi_E = \oint_s E.dA$

Electric Flux Density:

Electric flux per unit area.

Question

Find the electric flux crossing a imaginary cylinder placed in a uniform electric field



$$\phi_E = \oint_S E.dA$$

$$\phi_E = \oint_A E.dA + \int_B E.dA + \oint_C E.dA$$

$$\phi_E = E\int dA\cos\pi + E\int dA\cos 0 + E\int dA\cos(\pi/2)$$

$$\phi_E = -ES + ES = \underline{\underline{0}}$$

Gauss' Theorem:

The total electric flux Φ_E through any closed surface (Gaussian surface) with charge Q <u>inside</u> is equal to Q / ε_0

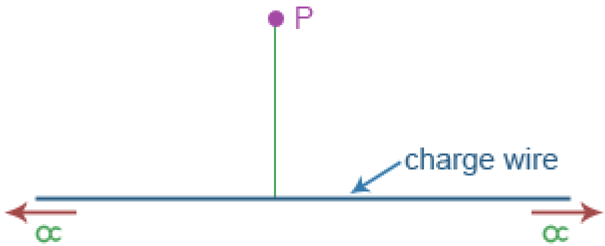
i.e. $\phi_E = Q / \varepsilon_0 \longrightarrow \oint E.dA = \dfrac{Q}{\varepsilon_0}$

Proof: Not given in this class. Yet if you are willing you can try !!!

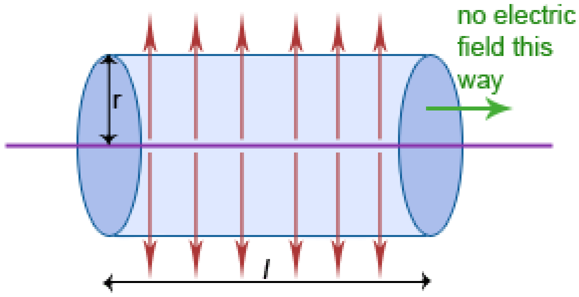
Applications of Gauss' Theorem:

(1) Consider an infinitely long wire. Find the electric field intensity at point P which is *r* away from the wire? The charge per unit length of the wire is λ.

Step 1. Draw a clear diagram



Step 2. Select a suitable closed surface (Gaussian surface)



$$\phi_E = \oint E.dA = Q/\varepsilon_0$$

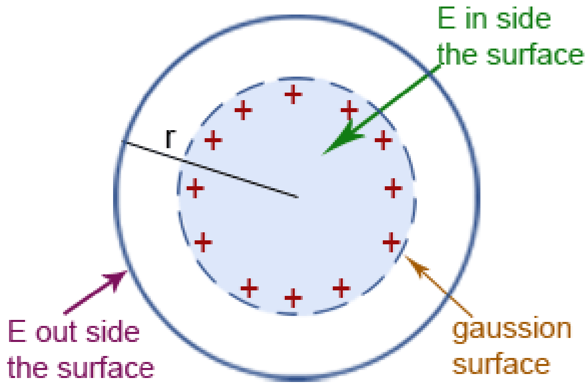
$$\phi_E = \oint E.dA = E.2\pi rl = l\lambda / \varepsilon_0$$

Step 3. Apply the Gauss' theorem

$$\Rightarrow E2\pi rl = l\lambda / \varepsilon_0$$

$$E = \underline{\underline{\frac{\lambda}{2\pi\varepsilon_0 r}}}$$

(2) Spherical Shell (Charge Q)



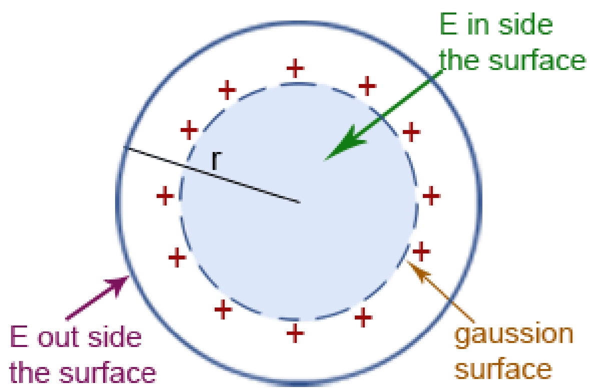
E inside the surface=0 ($\because Q = 0$)

$$\oint \underline{E}.\underline{dA} = Q / \varepsilon_0$$

E outside the surface $\longrightarrow$ $= 4\pi r^2 = Q / \varepsilon_0$

$$\Rightarrow E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

(3) Soild sphere (conductor) [Charge is distributed only on the surface]



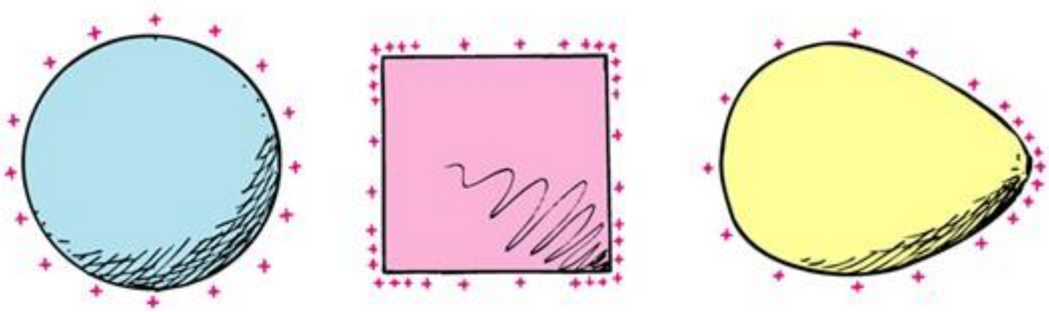
E inside the surface=0 ($\because Q = 0$)

$$\oint \underline{E}.\underline{dA} = Q / \varepsilon_0$$

E outside the surface $\qquad = 4\pi r^2 = Q / \varepsilon_0$

$$\Rightarrow E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

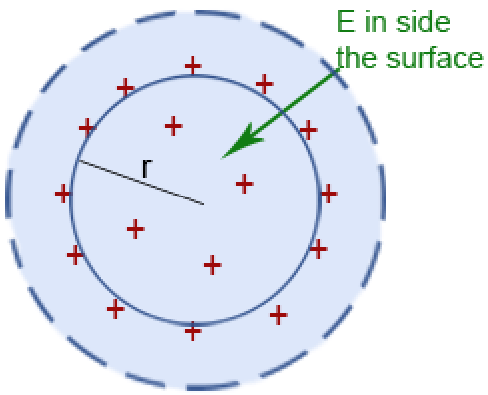
The electric field inside any charged conductor is zero- Electric Shielding…



What about gravitational field ?

Thinking Question: When lightning strikes a car, it is safe to sit inside it, because
A) The electric field is zero inside
B) The electric field is huge but only for a brief time
C) Nonsense! It is not safe – the electric field is huge inside the car

(4) Solid Sphere (insulator) [Charge is distributed all over]

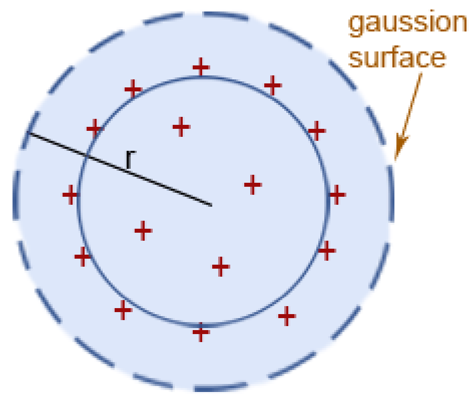


E inside the surface $\quad E_{in} 4\pi r^2 = q_{in} / \varepsilon_0$

If the charge density is ρ, the charge inside the Gaussian surface

$$= \frac{4}{3}\pi r^3 \rho$$

$$\Rightarrow E_{in} = \frac{4\pi r^3 \rho}{3\varepsilon_0 4\pi r^2} = \underline{\underline{\frac{\rho}{3\varepsilon_0} r}}$$

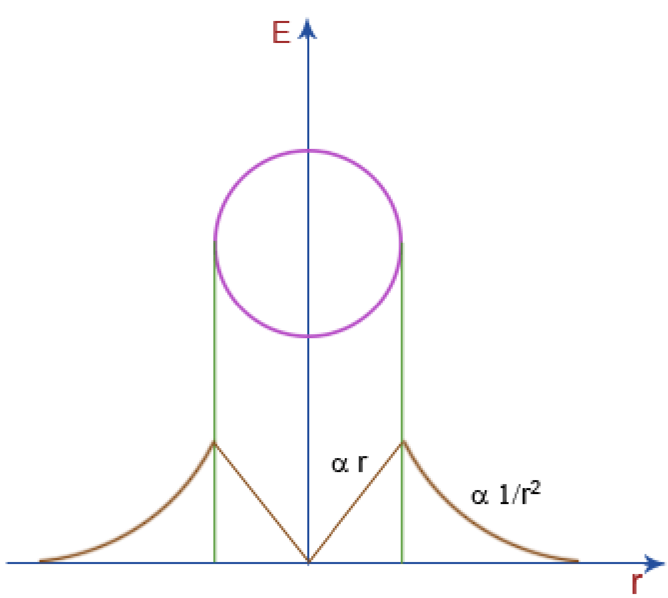


E outside the surface

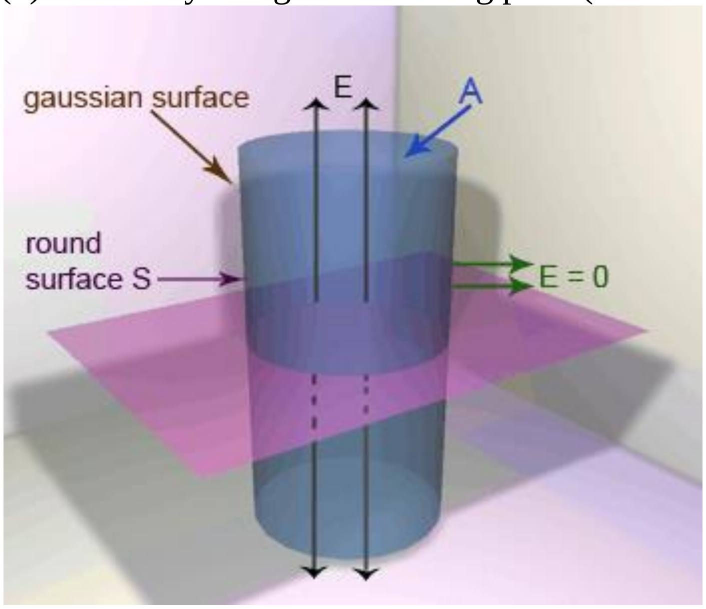
$$\Rightarrow E_{out} 4\pi r^2 = Q / \varepsilon_0$$

$$E_{out} = \frac{Q}{4\pi\varepsilon_0 r^2}$$

Graphical representation



(5) Uniformly charged conducting plate (thickness=0)



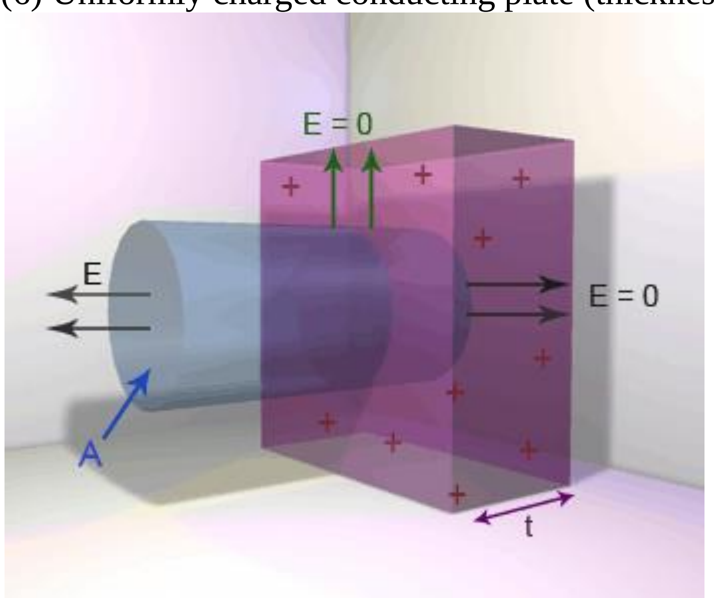
Charged density (surface) = $\sigma(cm^{-2})$

From Gauss' theorem

$$\phi = \frac{q}{\varepsilon_0}$$

$$(2A)E = \frac{A\sigma}{\varepsilon_0} \Rightarrow E = \frac{\sigma}{2\varepsilon_0}$$

(6) Uniformly charged conducting plate (thickness=t)



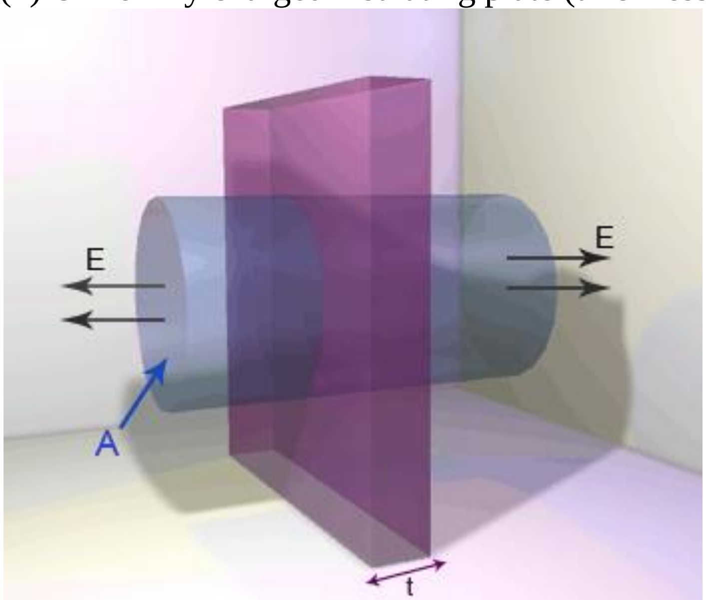
From Gauss' theorem

$$\phi = \frac{q}{\varepsilon_0}$$

$$AE = \frac{A\sigma}{\varepsilon_0} \Rightarrow E = \frac{\sigma}{\varepsilon_0}$$

σ- Surface charge density

(7) Uniformly charged insulating plate (thickness = t)



$$\phi = \frac{q}{\varepsilon_0}$$

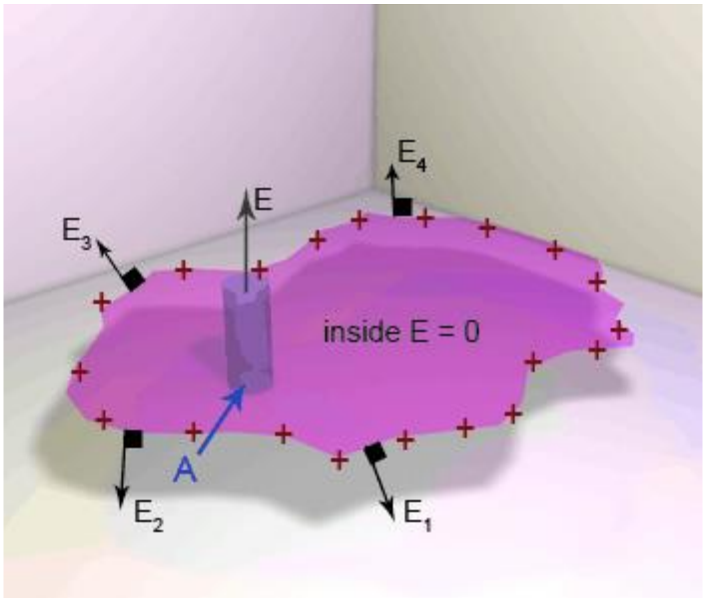
$$2AE = \frac{At\rho}{\varepsilon_0}$$

ρ- Volume charged density

$$\Rightarrow E = \frac{t\rho}{2\varepsilon_0}$$

(8) Conducting shell

- Inside a conducting shell there is electric shielding. -i.e. electric field is zero inside
- E is perpendicular to the surface
- E is strongest at those points where the net charge is heavily concentrated.



$$AE = \frac{A\sigma}{\varepsilon_0} \Rightarrow E = \frac{\sigma}{\varepsilon_0}$$

$$E_5 \rangle E_1$$